Bird Species Observation Analysis

By

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For Labmentix 4th Project

1. Executive Summary

This report presents a comprehensive analysis of bird species observations collected across forest and grassland habitats within 11 National Park Service administrative units in the eastern United States. The study, conducted during the 2018 breeding season (May-July), covers 17,077 individual observations of 127 unique species across 609 monitoring plots. The analysis was carried out in three stages:

- Python based data preprocessing
- SQL driven exploratory data analysis
- Power BI interactive dashboards

The central objective was to understand how habitat type, environmental conditions, time of day, and season, influence bird diversity and activity. Along with surface conservation relevant insights for land managers and policy makers.

Key Findings at a Glance

Dimension	Finding	Implication
Temporal	June peak (6,596 obs); dawn chorus 5–8 AM best window	Schedule surveys in June, 5–8 AM
Spatial	Forest & Grassland equally split (50/50); MONO plots richest	Prioritise Monocacy units for monitoring
Species	Cardinalis cardinalis most abundant (1,160 obs)	Songbird-rich eastern deciduous habitat
Activity	Singing dominant (57.8%); Early Morning = 79.9% of obs	Acoustic monitoring ideal for coverage
Environment	Light wind + partly cloudy = optimal; serious disturbance −44% obs	Control field-day selection by weather
Conservation	8 PIF watchlist species; 23.3% of obs on stewardship species	Focus resources on flagged species

2. Business Use Cases

Use Case	Description	Primary Beneficiary
Wildlife Conservation	Identify habitat hotspots and at-risk species for targeted protection	Park rangers, NGOs
Land Management	Optimise habitat restoration by matching species preferences to conditions	NPS land managers
Eco-Tourism	Map bird-rich plots for guided birdwatching tours in high-activity zones	Tourism boards
Sustainable Agriculture	Advise on buffer zones around grassland plots with high stewardship species	Agricultural planners
Policy Support	Supply evidence base for conservation regulations and funding allocation	Environmental agencies
Biodiversity Monitoring	Track population health trends across repeat visits and seasons	Ecologists, researchers

3. Dataset Description

3.1 Administrative Units

Code	Park / Site	Habitat
ANTI	Antietam National Battlefield	Forest & Grassland
CATO	Catoctin Mountain Park	Forest
CHOH	Chesapeake and Ohio Canal NHP	Forest & Grassland
GWMP	George Washington Memorial Parkway	Forest
HAFE	Harpers Ferry NHP	Forest & Grassland
MANA	Manassas National Battlefield Park	Forest & Grassland
MONO	Monocacy National Battlefield	Forest & Grassland
NACE	National Capital East Parks	Forest
PRWI	Prince William Forest Park	Forest
ROCR	Rock Creek Park	Forest
WOTR	Wolf Trap National Park	Forest

3.2 Key Columns

Column	Type	Description
admin_unit_code	Text	Park code (e.g., ANTI, MONO)
plot_name	Text	Unique monitoring point identifier
location_type / habitat	Text	Forest or Grassland
date / year / month	Date/Int	Observation date components
start_time / end_time	Time	Survey window timing
observer	Text	Field observer name
visit	Int	Visit number within season (1–3)
id_method	Text	Singing / Calling / Visualization
distance	Text	<= 50 m or 50–100 m from observer
flyover_observed	Bool	Bird flying over vs. perched
scientific_name / common_name	Text	Species identification
sex	Text	Male / Female / Undetermined
temperature / humidity	Float	Environmental conditions at survey time
sky / wind / disturbance	Text	Weather and disturbance conditions
pif_watchlist_status	Bool	On Partners in Flight watchlist
regional_stewardship_status	Bool	Regional conservation priority

4. Data Cleaning and Preprocessing

All preprocessing was performed in Python using the Pandas library in a Google Colab notebook. The pipeline followed a structured sequence from raw multi sheet Excel files to a single clean CSV ready for SQL import.

4.1 Preprocessing Pipeline

- Read all 11 sheets from each Excel workbook using pd.read_excel(sheet_name=None)
- Concatenated all sheets into a single DataFrame per habitat using pd.concat()
- Standardised column names: stripped whitespace and converted to lowercase
- Added a 'habitat' column ('Forest' or 'Grassland') to track origin
- Merged both habitat DataFrames into one combined dataset
- Removed duplicate rows with drop_duplicates()
- Handled missing values: sex – 'Unknown', flyover_observed – False
- Parsed data column to datetime – extracted year and month as integer features
- Exported the final cleaned dataset to bird_cleaned_data.csv

4.2 Schemas Differences Handled

- The forest and grassland Excel files had slightly different column sets:

Column	Present In	Handling
site_name	Forest only	Retained; NULL for Grassland rows
npstaxoncode	Forest only	Retained; NULL for Grassland rows
taxoncode	Grassland only	Retained; NULL for Forest rows
previously_obs	Grassland only	Retained; NULL for Forest rows

4.3 Data Quality Results

Check	Before Cleaning	After Cleaning
Duplicate rows	Present (combined concat)	Removed via drop_duplicates()
Missing sex values	~49% null	Filled with 'Unknown'
Missing flyover values	Some null	Filled with False
Column name consistency	Mixed case, spaces	All lowercase, underscore-separated
Date type	Mixed string/datetime	Parsed to datetime; year/month extracted
Habitat label	Absent	Added as explicit 'Forest' / 'Grassland' column

After cleaning, the combined dataset contained 17,077 rows and 34 columns with no duplicate records and clearly labelled habitat sources.

5. Exploratory Data Analysis

All EDA queries were written in MySQL and executed against the bird_cleaned_data table. Results are summarized below by analysis domain.

5.1 Temporal Analysis

Monthly Trends

Month	Total Observations	% of Total	Unique Species
May	5,596	32.8%	71
June	6,596	38.6%	69
July	4,885	28.6%	56

June had the highest observation volume (6,596) while May had the highest species richness (71 species). July saw a decline in both metrics, likely due to post breeding dispersal and reduced vocal activity.

Seasonal Trends

Season	Total Observations	Species Count
Monsoon (Jun–Aug)	11,481	114
Summer (Mar–May)	5,596	107

The Monsoon season (June-July in this dataset) accounted for 67.2% of all observations and supported 114 species, making it the richest survey window.

Hourly Activity (Observation Time)

Hour	Observations	Period
5:00	1,363	Early Morning
6:00	4,171	Early Morning
7:00	4,633	Early Morning — Peak
8:00	3,477	Early Morning
9:00	2,220	Late Morning
10:00	1,213	Late Morning

Hour 7 (7:00 – 8:00 AM) was the peak activity window with 4,633 observations. Early Morning (5 – 8 AM) accounted for 79.9% of all observations, reflecting dawn chorus effect in birds.

Scheduling all monitoring surveys to begin at or before 7:00 AM would capture nearly 80% of bird activity. Surveys starting after 9:00 AM miss the majority of vocalising birds.

5.2 Spatial Analysis

Observations by location types

Location Type	Observations	% Share	Unique Species
Forest	8,546	50.0%	108
Grassland	8,531	50.0%	107

Forest and Grassland habitats showed near-equal observation totals and species richness, indicating comparable survey effort and biodiversity. Forest had a marginal1-species edge (108 vs 107).

Top Plots by Total Observations

Rank	Plot Name	Observations	Admin Unit
1	MONO-0054	204	Monocacy NB
2	MONO-0094	118	Monocacy NB
3	MANA-0076	116	Manassas NBP
4	MONO-0102	88	Monocacy NB
5	MONO-0050	77	Monocacy NB

Biodiversity Hotspots (Species Richness per Plot)

Plot Name	Unique Species	Admin Unit
ANTI-0105	27	Antietam NB
MANA-0047	27	Manassas NBP
MONO-0057	27	Monocacy NB
CHOH-0812	26	C&O Canal NHP
MANA-0048	26	Manassas NBP

The Monocacy National Battlefield (MONO) dominates both observation volume and species richness rankings. It should be designated a priority conservation zone for bird monitoring.

5.3 Species Analysis

Top 10 most Observed Species

Rank	Scientific Name	Common Name	Observations
1	Cardinalis cardinalis	Northern Cardinal	1,160
2	Thryothorus ludovicianus	Carolina Wren	1,011
3	Vireo olivaceus	Red-eyed Vireo	739
4	Baeolophus bicolor	Tufted Titmouse	728
5	Passerina cyanea	Indigo Bunting	635
6	Contopus virens	Eastern Wood-Pewee	574
7	Spizella pusilla	Field Sparrow	542
8	Sturnus vulgaris	European Starling	522
9	Turdus migratorius	American Robin	496
10	Melanerpes carolinus	Red-bellied Woodpecker	493

Identification Method Distribution

ID Method	Observations	% of Total
Singing	9,863	57.8%
Calling	4,061	23.8%
Visualization	3,151	18.5%

57.8% of birds were identified by song, making acoustic monitoring the dominant and most efficient detection method. Visual identification contributed only 18.5%, confirming the importance of trained observers capable of audio identification.

Sex Distribution

Sex	Count	% of Total
Undetermined	8,410	49.3%
Male	3,338	19.5%
Female	146	0.9%
Unknown (filled)	5,183	30.3%

Female birds are dramatically underrepresented (146 confirmed) compared to males (3,338). This reflects typical field observation bias. Breeding season males are conspicuously vocal and colourful, while females remain cryptic and silent on nests.

Observation Interval Length

Interval	Observations	% of Total
0–2.5 min	8,605	50.4%
2.5–5 min	3,408	19.9%
5–7.5 min	2,705	15.8%
7.5–10 min	2,359	13.8%

Half of all detection occurred in the first 2.5 minutes of a survey point, suggesting that initial bird activity upon observer arrival is highest. This underscores the value of the point count methodology used.

5.4 Environmental Conditions

Sky conditions vs Observations

Sky Condition	Observations	% Share
Partly Cloudy	7,145	41.8%
Clear or Few Clouds	5,809	34.0%
Cloudy/Overcast	3,097	18.1%
Fog	647	3.8%
Mist/Drizzle	379	2.2%

Wind Conditions vs Observations

Wind Condition	Observations
Light air movement (1–3 mph)	8,511
Calm (< 1 mph)	4,417
Light breeze (4–7 mph)	3,737
Gentle breeze (8–12 mph)	412

75.8% of observations occurred under calm or light air conditions. Gentle breezes (8-12 mph) were associated with only 412 observations, likely because wind masks bird vocalisations and reduces detectability.

Disturbance Effects on Observations

Disturbance Level	Observations	Species Count
No effect on count	8,029	110
Slight effect on count	6,750	108
Moderate effect on count	1,819	90
Serious effect on count	479	62

As disturbance severity increased from 'No effect' to 'Serious effect', species count dropped from 110 to 62 (a 43.6% decline). Observation volume collapsed from 8,029 to 479 under serious disturbance conditions.

Disturbance is a major suppressor of both observation counts and species richness. Future surveys should flag seriously disturbed sessions and exclude them from trend analyses to avoid downward bias.

5.5 Distance and Behavior Analysis

Distance Band	Observations	% Share	Unique Species
<= 50 Meters	7,352	47.1%	~equal
50–100 Meters	8,239	52.9%	~equal

More observations occurred at 50–100 metres (52.9%) than within 50 metres, which is expected in dense habitat where closer birds may be obscured by vegetation. Species richness was comparable across both bands.

Flyover behavior

Flyover Observed	Observations	% of Total
False (Perched/Singing)	15,591	91.3%
True (Flying Over)	1,486	8.7%

91.3% of observations were of perched or singing birds, with only 8.7% classified as flyovers. Flyover species included predominantly aerial foragers such as swallows, swifts, and raptors.

5.6 Observer Trends

Observer	Total Observations	Unique Species	Avg Obs/Visit
Elizabeth Oswald	6,391	120	~5,690
Kimberly Serno	5,902	91	~5,690
Brian Swimelar	4,784	84	~5,690

Elizabeth Oswald recorded both the most observations (6,391) and the highest species diversity (120 species). Brian Swimelar had the fewest observations (4,784) and lowest species count (84), which may reflect plot assignments, visit scheduling, or observer experience rather than bias per se.

Visit Patterns

Visit	Observations	Species Count
Visit 1 (May)	7,588	71
Visit 2 (June)	6,801	69
Visit 3 (July)	2,688	56

Visit 3 (July) saw a sharp 60% drop in observations compared to Visit 1, and a 15-species decline in richness. This is consistent with post-breeding season when territorial singing ceases and juveniles become mobile but cryptic.

5.7 Conservation Insights

PIF (Partners in Flight) Watchlist Status

Status	Observations	Unique Species
Not on Watchlist	16,699	119
On PIF Watchlist	378	8

Top 5 At-Risk Species by Observation Count

Scientific Name	Common Name	Observations	Habitat
Cardinalis cardinalis	Northern Cardinal	1,160	Both
Thryothorus ludovicianus	Carolina Wren	1,011	Both
Vireo olivaceus	Red-eyed Vireo	739	Both
Baeolophus bicolor	Tufted Titmouse	728	Both
Passerina cyanea	Indigo Bunting	635	Both

Regional Stewardship Status

Status	Observations	% of Total
Not a Stewardship Priority	13,092	76.7%
Regional Stewardship Priority	3,985	23.3%

At-Risk Species by Habitat

Habitat	At-Risk Species Count
Forest	108
Grassland	107

8 species on the PIF Watchlist were detected, accounting for 378 observations. Forest and Grassland habitats supported nearly equal numbers of at-risk species (108 vs 107), highlighting the conservation value of both habitat types equally.

23.3% of all observations (3,985 records) involved species designated as Regional Stewardship Priorities, indicating that roughly one in four birds observed in this study warrants active conservation attention.

6. Power Bi Dashboards Overview

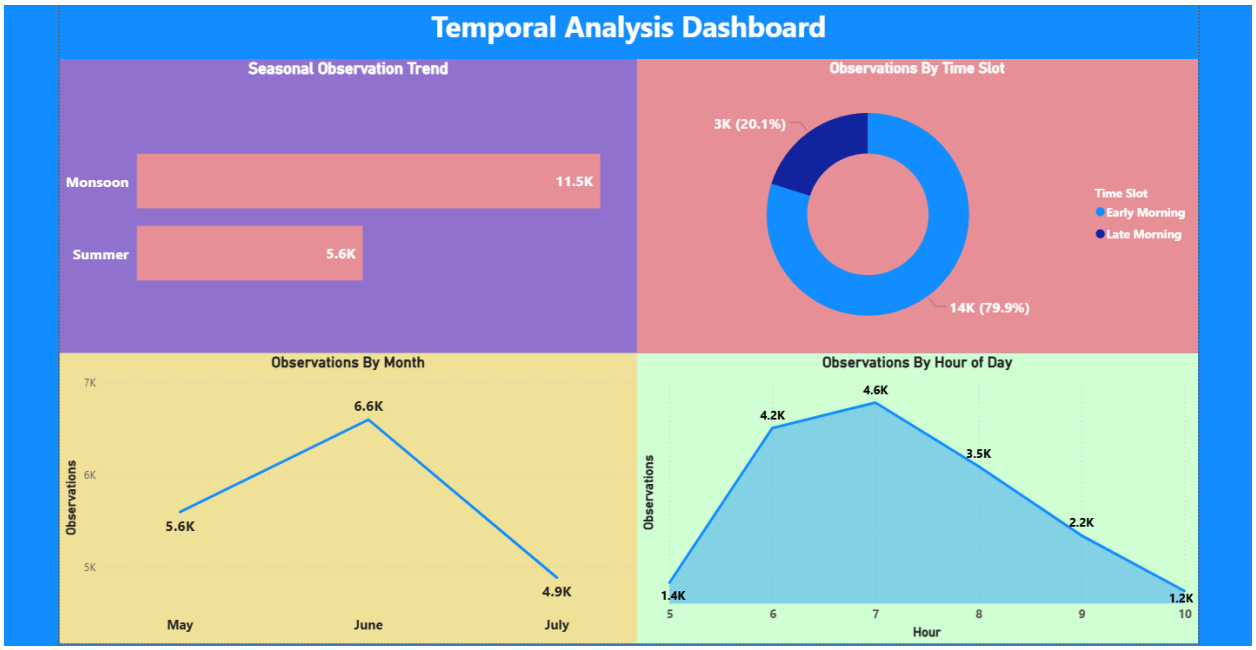
Seven Power BI dashboard pages were developed to visualise the EDA findings. Each page corresponds to a major analytical dimension, and some include slicers for habitat type, observer, and visit number.

Dashboard Page	Key Visuals	Primary Insight Delivered
Temporal Analysis	Seasonal bar chart, monthly line chart, hourly line chart, time-slot donut	Peak season (June/Monsoon) and dawn chorus window (7 AM)
Spatial Analysis	Donut by location type, bar by plot name, bar by species/plot, matrix table	MONO hotspot; equal Forest/Grassland richness
Species Analysis	KPI card (127 spp), bar by interval, donut by sex, donut by ID method, clustered bar M/F	Singing dominance; male observation bias
Environmental Analysis	Area chart (temp), line chart (humidity), donut (sky), bar (wind), bar (disturbance)	Light air + partly cloudy optimal; disturbance suppresses diversity
Distance & Behaviour	Bar by distance, donut species by distance, bar flyover, donut flyover %	91.3% perched; 50–100m band most common
Observer Analysis	Bar by observer, slicer (observer/visit), bar visits, stacked bar visit×habitat, matrix	Elizabeth Oswald highest diversity; Visit 3 sharp drop
Conservation Analysis	Bar by AOU code, donut watchlist, bar stewardship, matrix, bar at-risk by habitat	8 PIF watchlist spp; Forest/Grassland equally at risk

All dashboards were built directly from the MySQL query output CSVs, ensuring consistency between the SQL analysis and visual results. Slicers allow filtering by Observer, Visit, and Habitat Type across all relevant pages.

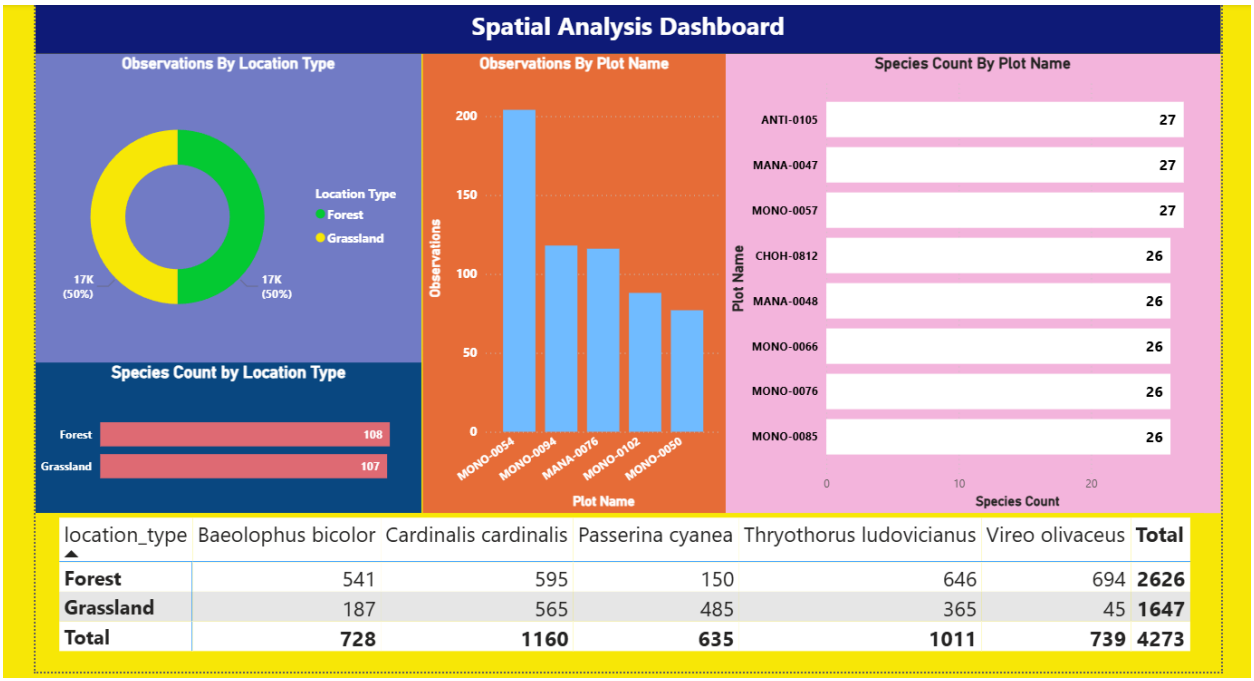
7. Key Findings and Actionable Recommendations

7.1 Temporal Recommendations



- Schedule all field surveys between 5:00–8:00 AM to capture the dawn chorus peak (79.9% of observations occur in this window).
- Prioritise June surveys over May and July — June yields 18% more observations than May and 35% more than July.
- Budget for at least 2 visits per season (Visit 1 + Visit 2) as Visit 3 shows a 60% observation decline.

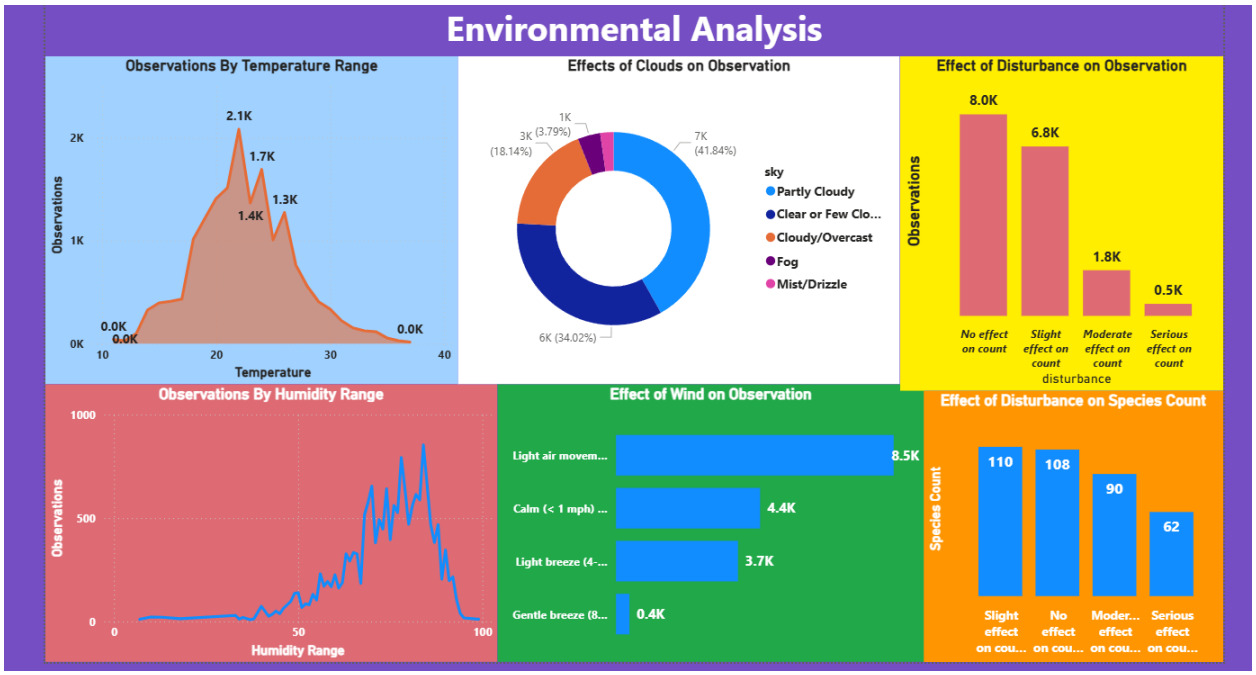
7.2 Spatial Recommendations



location_type	Baeolophus bicolor	Cardinalis cardinalis	Passerina cyanea	Thryothorus ludovicianus	Vireo olivaceus	**Total**
Forest	541	595	150	646	694	**2626**
Grassland	187	565	485	365	45	**1647**
Total	**728**	**1160**	**635**	**1011**	**739**	**4273**

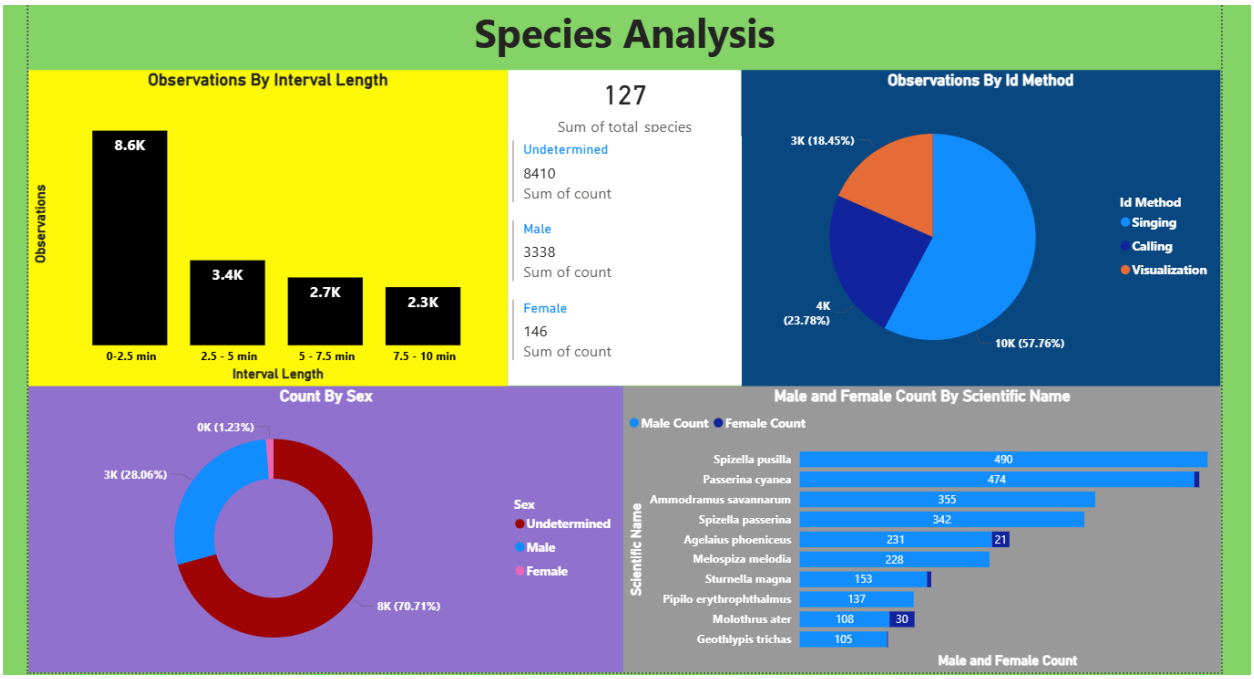
- Designate the Monocacy National Battlefield (MONO) as a Tier-1 biodiversity monitoring site, as it holds the most observation-rich and species-diverse plots.
- Maintain equal monitoring effort in Forest and Grassland habitats; both support 107–108 species and similar conservation-priority numbers.
- Focus species-specific conservation actions on plots ANTI-0105, MANA-0047, and MONO-0057, which have the highest species richness (27 species each).

7.3 Environmental Recommendations



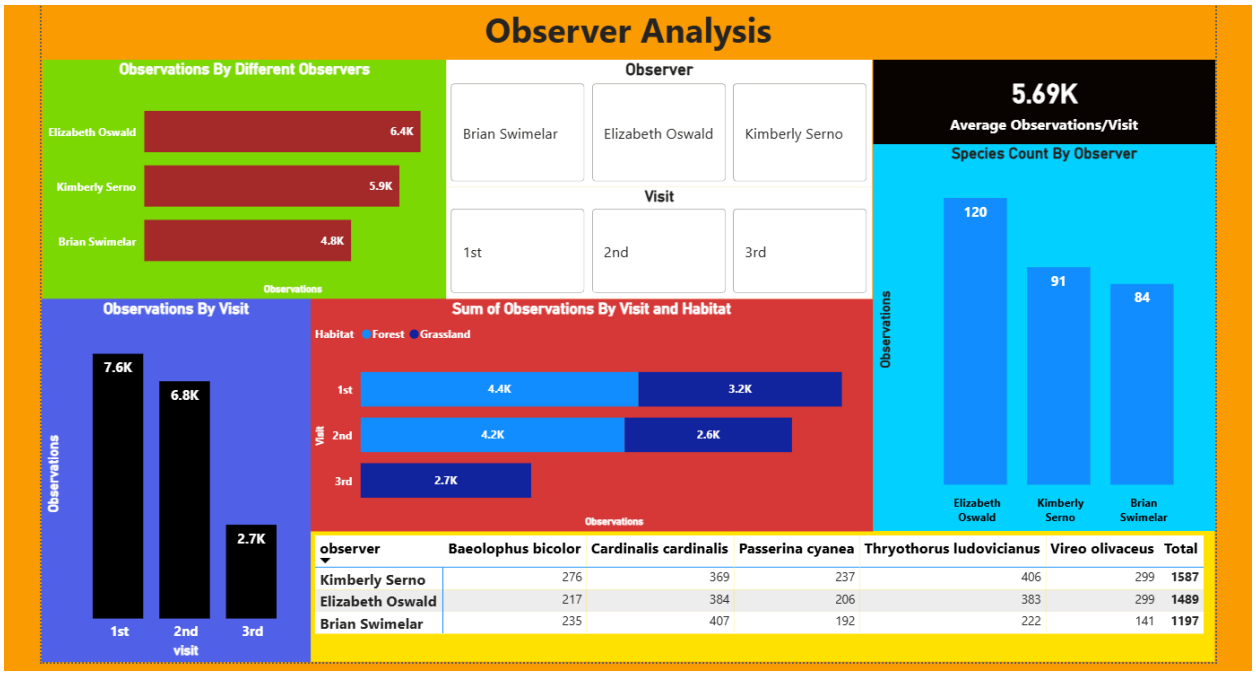
- Cancel or reschedule surveys when disturbance is classified as 'Moderate' or 'Serious', as these conditions suppress species richness by up to 43.6%.
- Avoid surveying in gentle-breeze conditions (8–12 mph+); conduct surveys only when wind is calm or light-air to maximise acoustic detection.
- Optimal survey conditions: Partly Cloudy sky + Light air movement + No disturbance.

7.4 Species Conservation Recommendations



- Target acoustic monitoring for the 8 PIF Watchlist species, particularly during May–June when vocal activity is highest.
- Investigate the 43-species gap between 'Most Observed' (Cardinalis cardinalis, 1,160) and median species counts — high dominance by a few species may indicate habitat degradation or observer bias.
- Commission a focused grassland study to understand the high proportion of undetermined-sex observations as cryptic female grassland species may be systematically undercounted.

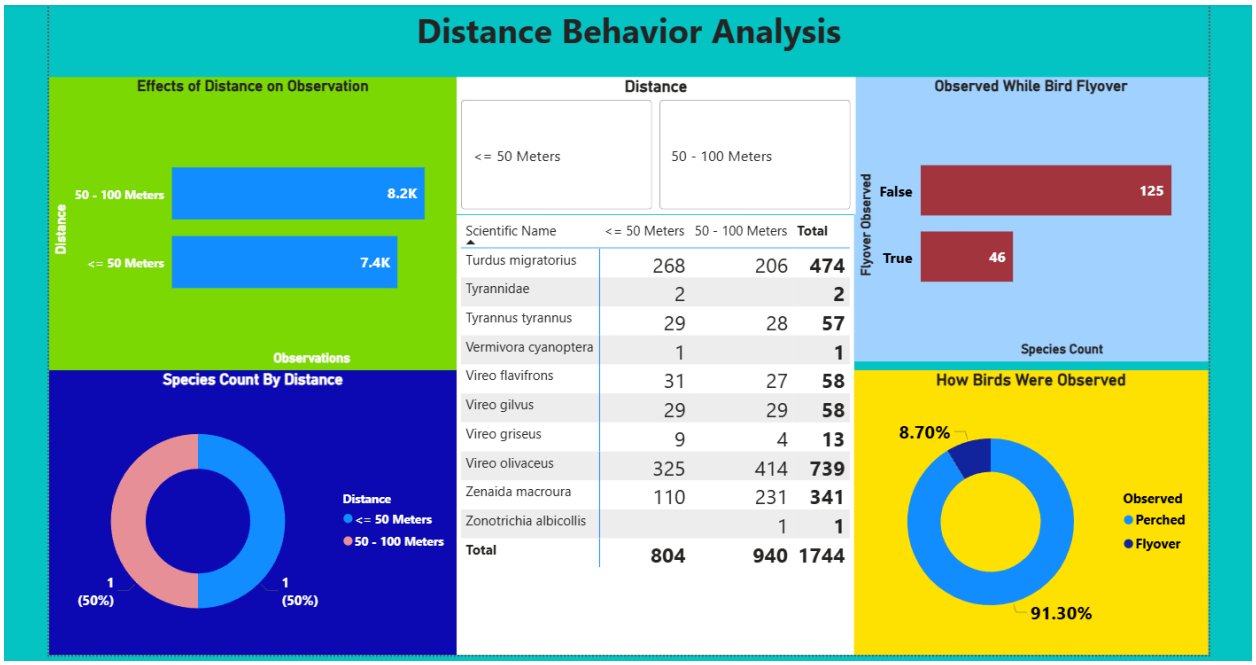
7.5 Observer and Methodology Recommendations



observer	Baeolophus bicolor	Cardinalis cardinalis	Passerina cyanea	Thryothorus ludovicianus	Vireo olivaceus	Total
Kimberly Serno	276	369	237	406	299	**1587**
Elizabeth Oswald	217	384	206	383	299	**1489**
Brian Swimelar	235	407	192	222	141	**1197**

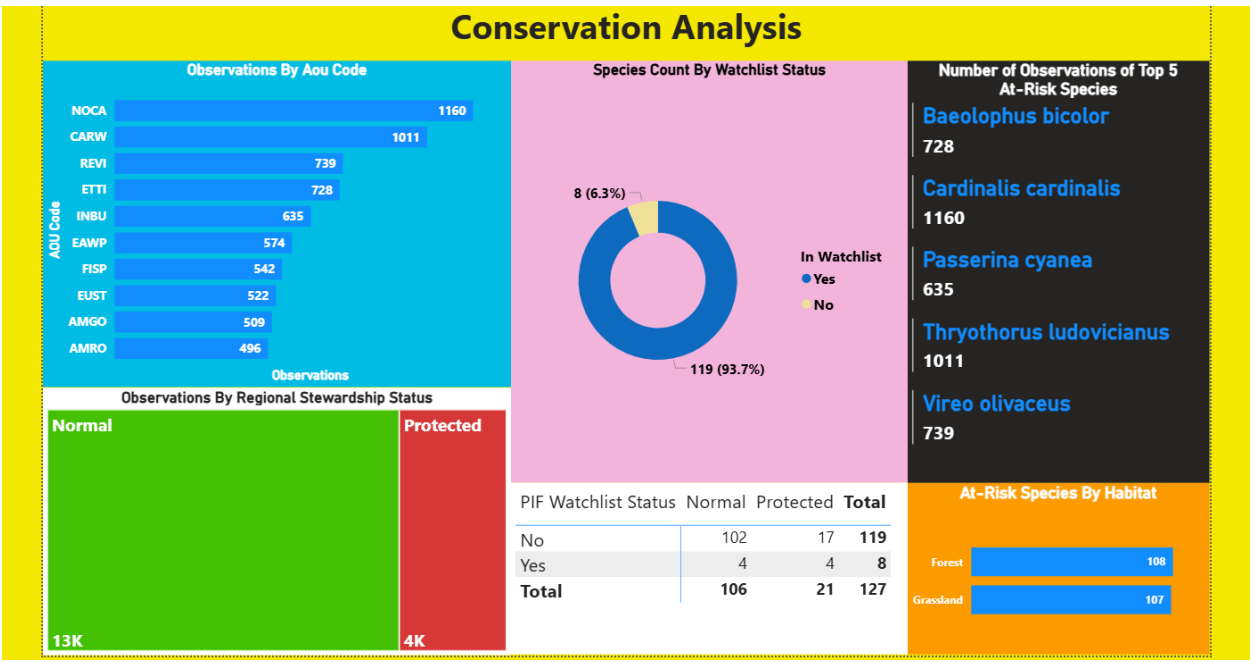
- Standardise plot assignments across observers to reduce observer-level confounding in species count comparisons.
- Implement rigorous training in audio identification to increase the proportion of Singing-based detections (currently 57.8%) and reduce reliance on Visualization (18.5%), which is harder to standardise.
- Consider deploying passive acoustic recorders at high-richness plots to supplement point-count surveys and capture species active outside the survey window.

7.6 Distance and Behavior Recommendations



- Develop species-specific "proximity zones" for field teams.
- For species like Vireo olivaceus, which are predominantly seen at greater distances (414 at 50-100m), observers should be equipped with higher-magnification optics to ensure identification accuracy doesn't drop as distance increases.
- Since flyover observations are rare but provide critical data on migration and movement, implement a "Rapid-ID" training module for field staff.
- This will help increase the success rate of identifying the 46 "True" Flyover observations seen in the bar chart, ensuring that fleeting encounters aren't lost to "unidentified" categories.

7.7 Conservation Insights Recommendations



PIF Watchlist Status	Normal	Protected	Total
No	102	17	119
Yes	4	4	8
Total	106	21	127

- Since 50% of at-risk species are being found in non-protected "Normal" areas, it would be beneficial to propose buffer zone extensions or new protected designations for the habitats where these "Normal" area watchlist species were recorded.
- Rather than focusing on one biome, they should adopt a dual-track conservation strategy.
- Forest: Focus on canopy-dwelling species like Vireo olivaceus (739 observations).
- Grassland: Focus on ground or low-brush species.
- Resource Allocation: Allocate 50% of the conservation budget to each habitat to maintain the high diversity levels currently observed in both.
- Use the "high-frequency" species as Indicator Species for the health of the local ecosystem.
- As they are easy to find and monitor, changes in their population or distance behavior (from the first dashboard) can serve as an early warning system for broader environmental shifts before rarer species are impacted.

8. Limitations and Future Work

8.1 Current Limitations

- Single year (2018): No multi-year trend analysis possible. A 3–5 year dataset would enable detection of population trends.
- Three observers only: Inter-observer reliability was not formally tested; observer-level species counts show material differences.
- No GPS coordinates: Spatial analysis is limited to plot names; geographic mapping and habitat quality scoring are not possible.
- Start/end times were partially missing: Hourly analysis is based on available records only and may not fully represent all survey effort.
- No abundance counts: Only presence/detection data; population size estimates are not possible with this dataset.

8.2 Recommended Future Work

- Multi-year expansion: Repeat the same pipeline with 2019–2024 data to identify declining species and emerging hotspots.
- Geospatial integration: Add GPS coordinates to plot names and build ArcGIS or QGIS habitat maps overlaid with species richness.
- Machine learning: Apply clustering (K-Means) to group plots by environmental conditions and species assemblage; use random forest models to predict species richness from weather variables.
- Acoustic data integration: Link point-count detections with passive acoustic recorder data from the same plots to validate detection rates.

9. Conclusion

This project successfully delivered an end-to-end bird species observation analysis pipeline from raw multi sheet Excel data through Python preprocessing, MySQL EDA, and Power BI visualisation, addressing all seven analytical domains specified in the project brief.

The analysis revealed that Forest and Grassland habitats support near-equal bird diversity (108 vs 107 species), that June mornings (5–8 AM) are the optimal survey window, that disturbance dramatically suppresses species richness, and that 23.3% of observations involve Regional Stewardship Priority species requiring active conservation management.

The Monocacy National Battlefield emerged as the most important biodiversity hotspot, and Elizabeth Oswald as the most productive observer by both volume and species diversity. The 8 PIF Watchlist species identified should be the focus of targeted future monitoring campaigns.

The interactive Power BI dashboards provide land managers, ecologists, and policymakers with an intuitive interface for exploring these findings, filtering by habitat, observer, and visit, and extracting data-driven decisions on conservation resource allocation.